

MATH0020 Curves and Surfaces

<i>Year:</i>	2025–2026
<i>Code:</i>	MATH0020
<i>Level:</i>	6 (UG)
<i>Normal student group(s):</i>	UG: Year 2 Mathematics degrees PG: (2026-27) MSc Mathematics
<i>Value:</i>	15 credits (= 7.5 ECTS credits)
<i>Term:</i>	2
<i>Assessment:</i>	90% examination, 10% coursework
<i>Normal Pre-requisites:</i>	MATH0011, MATH0003, MATH0005
<i>Lecturer:</i>	Prof C Bellettini

Course Description and Objectives

This course concerns the geometry of smooth curves and surfaces in $\mathbb{R}^3$. We will begin by looking at local properties, i.e., properties such as curvature, which are defined using a small neighbourhood of a point. We will go on to prove global results in which we study the curve or surface as whole. For example, the Gauss-Bonnet Theorem relates the geometry and topology of a surface. We will also study special surfaces such as minimal surfaces, which are natural models for soap films.

Recommended Texts

Manfredo Do Carmo, *Differential Geometry of Curves and Surfaces* (Prentice Hall).

Detailed Syllabus

Curves: Review of curvature and torsion of curves and Frenet-Serret formulae, isoperimetric inequality, total curvature and the global geometry of curves.

Surfaces in $\mathbb{R}^3$: First fundamental form, length and area, normal and geodesic curvature, geodesics, second fundamental form, the Gauss map, curvature (principal, Gaussian, mean, geodesic and normal), Theorema Egregium, Gauss-Bonnet Formula, Euler characteristic, minimal surfaces, applications.